

DIM3: Display as an Interface Medium to Three-dimension with Swept a Bending-Plastic-Plate

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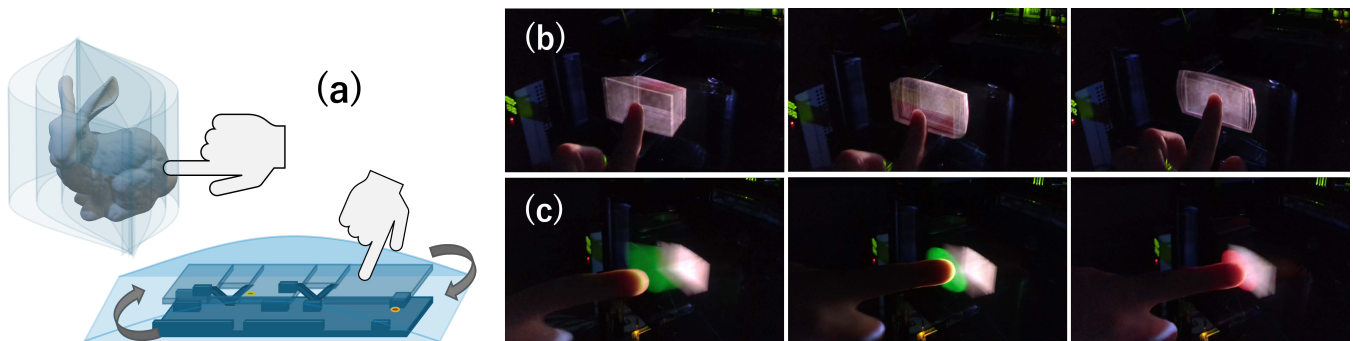


Figure 1: Direct and physical interaction with virtual objects. (a) DIM3 concept. (b) Interactive application where a rectangular prism deforms as if being crushed. (c) Application where the button color changes from green to red based on the depth of the press.

Abstract

Existing volumetric displays do not allow safe bare-hand touch of virtual objects, and even when a hand can enter the volumetric space, force feedback from the virtual objects is not provided. Consequently, real-world-like physical interaction has not yet been achieved. In this demonstration, we propose DIM3, which enables bare-hand touch and manipulation of virtual objects while providing force feedback at the point of touch, and we introduce a novel interaction technique enabled by it. DIM3 employs a swept-volume approach by driving a single plastic plate with moderate bending elasticity in reciprocating motion while bending it, allowing safe contact and force feedback at the touched location. Moreover, DIM3 detects the finger's depth position via intermittent contact between the screen and the finger, enabling virtual object manipulation in which the input and output spaces coincide.

CCS Concepts

• Human-centered computing → Interaction devices; Displays and imagers; Mixed / augmented reality.



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Keywords

Volumetric Display, Swept Bending Plate, Direct Touch Interaction, Force Feedback, High-Speed Projection, Spatial Augmented Reality

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1 Introduction

In contexts such as sharing product images in stores and business meetings and public signage, where three-dimensional structures must be understood and shared among multiple people, there is strong demand for 3D displays that can be directly touched and manipulated. In particular, because holographic and volumetric displays can reproduce depth cues such as binocular disparity, motion parallax, and accommodation without auxiliary devices such as goggles, users expect to manipulate rendered virtual objects with bare hands like real objects and to receive force feedback from the virtual objects.

Holographic displays generate stereoscopic hologram images by reproducing the object-light wavefront with a spatial light modulator. To add haptics, combining holographic displays with mid-air

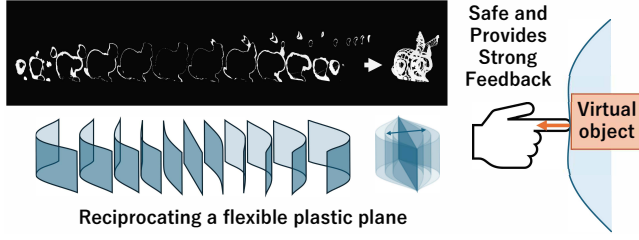


Figure 2: The bending screen produces a persistence-of-vision effect. Users can touch it without injury, and it provides strong force feedback.

haptic devices has been proposed [4, 8], but the pressure produced by ultrasound arrays or compressed air [11] is weak and insufficient for force feedback.

Volumetric displays render 3D imagery by physically placing light points in space, enabling 360-degree viewing by multiple users. In volumetric scanning methods [5, 6, 10, 13, 14], 2D cross-sections are sequentially projected onto a rapidly reciprocating or rotating optical diffuser to form a volumetric image via persistence of vision, but the diffuser has traditionally been rigid, preventing bare-hand contact. Some approaches allow hand insertion by replacing the diffuser with parallel elastomer bands [3] or by filling the volume with light-diffusing media [1, 2, 7, 12], yet force feedback remains unrealized. A femtosecond-laser air-plasma approach [9] provides touchable plasma and tactile sensation, but contact is limited to extremely short durations for safety and imagery is restricted to monochrome. Therefore, direct touch of virtual objects with physical interaction, as users would expect, has not yet been realized.

In this study, we propose DIM3, a volumetric display based on volumetric scanning that drives a single plastic plate with moderate bending elasticity in reciprocating motion while bending it. By mitigating impacts to the finger through the screen's moderate bending elasticity, this method enables users to safely touch virtual objects, and by driving the screen so as to bend, it realizes strong force feedback. DIM3 detects the fingertip's depth position by leveraging intermittent contact between the screen and the finger. Accordingly, we aim to realize direct and physical interaction with stereoscopic images, as shown in Fig.1(a). In this demonstration, we present an interaction in which the image deforms and changes in response to touch and press depth.

2 DIM3

DIM3 is a swept volumetric display that enables touch input and force feedback. DIM3 drives a single screen in reciprocating motion while bending it, sequentially projects cross-sections of a 3D object, and renders a 3D image through the persistence-of-vision effect (Fig.2). By using a single thin plastic plate as the screen material, the screen deforms upon contact with the finger and absorbs impact, ensuring safe contact with the virtual object. Because a non-stretchable plastic plate is driven so as to bend, a strong restoring force acts in the direction that pushes the finger back even after contact. Accordingly, the user receives force feedback at the touched location, as if being pushed back by the virtual object.

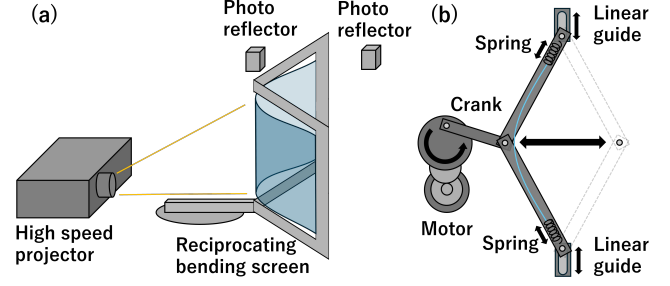


Figure 3: System Configuration. (a) Layout of the main hardware. (b) Reciprocating Mechanism Operation.

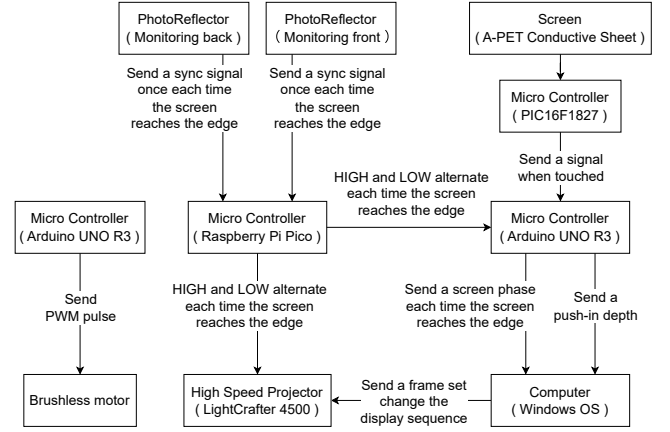


Figure 4: System Components.

3 Implementation

There are two possible implementation approaches for the proposed method: using a flexible display or using a projector. Using a flexible display requires durability against repeated high-speed bending and stretching, impact resistance against contact by a finger or similar objects, and a high refresh rate, but no commercially available flexible display satisfies all of these requirements. Therefore, in this study, we adopt the high speed projector based approach.

3.1 Reciprocating mechanism

An overview and operation of the reciprocating mechanism are shown in Fig.3. A motor is PWM-controlled by an Arduino UNO R3 to drive a crank that scans a single screen at 8 Hz in reciprocating motion while bending it. Because bending changes the distance between the screen ends and can cause instantaneous rebound, compression springs are used to absorb this variation. Two photoreflectors placed at the front and rear detect the screen's turning points. To enable interaction with the virtual object, the screen also functions as a self-capacitance touch sensor. As the screen material, we use an A-PET conductive sheet with a CNT coating, which is resistant to bending and has impact resistance. Because A-PET has high transparency and is not suitable for rear projection, the back side of the sheet is painted black using a soft vinyl paint that forms an elastic coating film. The screen size is 65 mm in height and 145

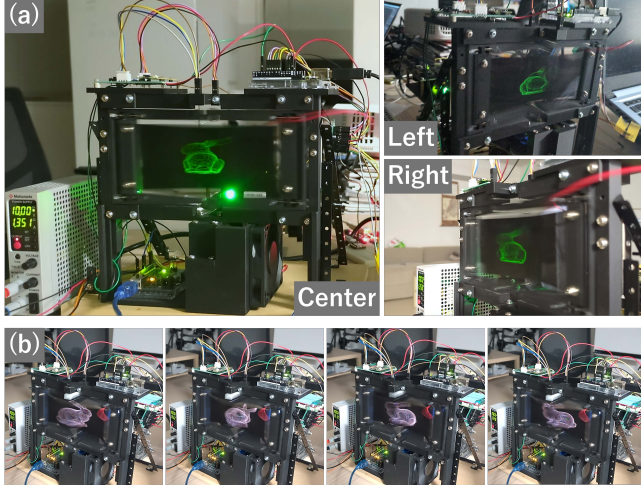


Figure 5: 3D Rendering with DIM3. (a) Motion parallax when drawing a stationary Stanford-bunny. (b) Animation of a rotating Stanford-bunny.

mm in width, and the display volume during reciprocating motion is a spindle-shaped prismatic volume with a depth of 40 mm.

3.2 Rendering system

To prevent the image from being occluded by the user's hand, the image is projected from the rear. The system configuration is shown in Fig.4. The output images are generated as follows. First, 90 cross-sectional images are created from the 3D model to be displayed, along the screen's scanning direction. These cross-sectional images are projected at 1440 Hz using a high-speed projector (LightCrafter4500). Due to device constraints, the projected images are binary images with a resolution of 912×1140 pixels. The computer for projection timing receives a signal indicating the screen's turning points. It then projects the 90 sliced cross-sectional images in an appropriate order synchronized to the signal. Turning-point timing is relayed by a Raspberry Pi Pico to the high-speed projector and an Arduino UNO R3. Touch is detected by the PIC16F1827 Capacitive Sensing module and sent to the Arduino UNO R3. The press depth is estimated by first calculating the screen position at the instant of touch from the elapsed time between the moment when the screen turns at the far side and the moment when touch is detected.

3.3 Implementation results and evaluation

Fig.5(a) shows the implementation results. 3D images are rendered by the implemented volumetric display and can be observed from all directions except those physically occluded by mechanical components such as the driving parts, which provides motion parallax. We confirmed that users can directly touch the virtual objects with bare hands, safely manipulate them, and perceive the reaction force from the virtual objects. We then conducted the following exploratory evaluations focusing on this perceptible reaction force.

3.3.1 Evaluation 1: Press depth and reaction force. The reaction force presented by our method varies with the screen's elastic force

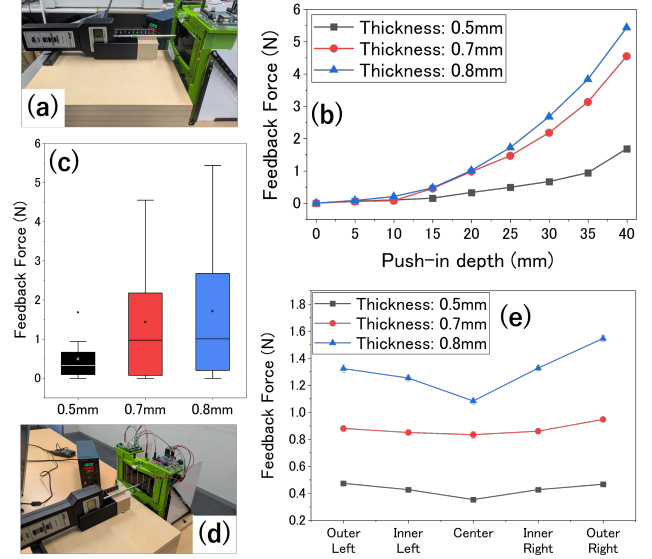


Figure 6: (a) Experimental setup for Evaluation 1. (b) Reaction force vs. push-in depth. (c) Box plot of reaction force vs. push-in depth. (d) Experimental setup for Evaluation 2. (e) Reaction force vs. push position.

and its velocity at the moment of collision. The experimental setup and results are shown in Fig.6(a-c). To characterize this dependency, we investigate how reaction force changes with press depth for screens made of A-PET conductive sheets with thicknesses of 0.5, 0.7, and 0.8 mm. Measurements are taken at the screen center using a digital force gauge, where the motion amplitude is largest. Press depth is set from 0 to 40 mm in 5 mm increments, and each condition is measured three times to compute the mean. The experiment was conducted as follows: (1) A digital force gauge was mounted on a stage and aligned perpendicular to the screen with its measurement axis at the screen center. (2) The display was activated and the screen was reciprocated at 8 Hz. (3) The gauge was slowly slid to press the screen to the target depth, held for 3 seconds, and the maximum reaction force was recorded.

We conducted a Two-way ANOVA (including interaction) with reaction force as the dependent variable and thickness (0.5/0.7/0.8 mm) and press depth (0–40 mm, 8 levels) as factors. The results showed a significant main effect of thickness ($F(2, 48) = 4552.76$, $p < 0.01$, $\eta p^2 = 0.995$). The effect of press depth was also significant ($F(7, 48) = 6045.53$, $p < 0.01$, $\eta p^2 = 0.999$). In addition, the interaction between thickness and press depth was significant ($F(14, 48) = 574.36$, $p < 0.01$, $\eta p^2 = 0.994$).

3.3.2 Evaluation 2: Contact location and reaction force. The screen reciprocates while bending, the reaction force presented to the fingertip varies with contact location. Therefore we measure the reaction force for screens made of A-PET conductive sheets with thicknesses of 0.5, 0.7, and 0.8 mm at five locations (center and ± 24 mm and ± 48 mm from center). To enable comparison across locations with different pressable depths, we normalize the press depth at each location by half of the local motion amplitude. The

local motion amplitude is measured using a digital force gauge. Each condition is measured three times to compute the mean. The amplitude is 40 mm at the center, 36 mm at the inner positions, and 20 mm at the outer positions. The experiment was conducted as follows: (1) A digital force gauge was mounted on a stage and aligned perpendicular to the screen with its axis at the measurement location. (2) The display was activated and the screen was reciprocated at 8 Hz. (3) The gauge was slid to press the screen to the target depth, held for 3 seconds, and the maximum reaction force was recorded.

The experimental setup and results are shown in Fig.6(d,e). For each thickness condition, we conducted a One-way ANOVA with press location as the factor. As a result, the main effect of press location was significant for all thickness conditions (0.5 mm: $F(4, 10) = 93.27$, $p < 0.01$; 0.7 mm: $F(4, 10) = 15.55$, $p < 0.01$; 0.8 mm: $F(4, 10) = 150.45$, $p < 0.01$). We collapsed symmetric left–right locations into three levels (outer/inner/center). We then performed post-hoc comparisons using paired t-tests with Bonferroni correction (significance level $\alpha = 0.0167$). For the 0.5 mm thickness, all comparisons were significant: outer–inner, outer–center, and inner–center. The reaction force increased toward the outer position. For the 0.7 mm thickness, outer–inner and outer–center were significant, whereas inner–center was not significant. This confirms that the reaction force increased at the outer position. For the 0.8 mm thickness, all comparisons were significant: outer–inner, outer–center, and inner–center. As in the 0.5 mm condition, the reaction force increased toward the outer position.

3.4 Discussion

The system provides a perceptible increase in reaction force as the user presses deeper while directly touching the moving screen. The results of evaluation showed that for the 0.7 mm and 0.8 mm screens, when the press depth exceeded 20 mm, the finger was pushed back. This indicates that the force feedback is sufficient for interaction.

Evaluation 1 showed that reaction force increased with thickness and press depth, and that the thickness-related force difference depended on press depth. A possible reason why the reaction force increased with press depth is that the screen bent more, which increased the restoring force. This suggests potential applications such as replicating the behavior of a palm, where the surface feels soft but fingers do not penetrate deeper layers.

In Evaluation 2, the reaction force varied significantly with press location. In particular, a consistent tendency was observed across thickness conditions: the reaction force was larger at the outer locations. A possible reason is that the screen is driven with its two ends as pivots. Therefore, the closer the contact location is to a pivot, the larger load is required to deform the screen. In this implementation, the amplitude is maximal at the center; however, mechanically offsetting the synchronization of the screen's two ends can shift the location of maximum amplitude, which may enable dynamic modulation of the presented reaction force. Note that, in this analysis, the number of samples per thickness condition is 1, and the repetitions are technical replicates. Therefore, the inferential statistics primarily reflect the magnitude of differences relative to measurement reproducibility (measurement error). To generalize

the thickness effect, further validation should be conducted with a larger number of independent samples for each thickness condition.

4 Demonstration

Fig.5 shows the demonstration in which DIM3 renders 3D imagery, including moderately complex 3D objects and low-frame-animations such as rotation. Because multiple viewers can simultaneously perceive the 3D imagery from all directions with naked-eye viewing, DIM3 can be broadly used for digital signage in public spaces. Examples include 3D advertisements, maps, and exhibit explanations in museums.

Fig.1(b,c) shows demonstrations in which users directly touch virtual objects to deform or change them. Users can press a rendered cuboid along the depth direction while perceiving a repulsive force, and can also press a rendered button whose color changes from red to green at a certain depth. DIM3 enables these force-feedback interactions without gloves or additional cameras for hand tracking, making it suitable as a 3D design tool for intuitive manipulation and as a portable 3D input interface.

5 Conclusion

In this demonstration, we proposed DIM3, a volumetric display that drives a single plastic plate with moderate bending elasticity in reciprocating motion while bending it. DIM3 allows users to directly touch virtual objects to deform or change them, and presents strong force feedback at the touched location. It detects the fingertip's depth position via intermittent contact between the screen and the finger, and thus does not require conventional bulky camera-based hand tracking. Evaluation experiments confirmed that users can directly touch virtual objects with bare hands and that the system can present reaction forces sufficient for interaction. This approach has potential for digital signage in public spaces and wearable 3D input/output interfaces, and it is expected to support a wide range of applications.

Acknowledgments

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